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Thursday 24 September 2026

Associate Professor Scott Barnett, Stefanus Kurniawan
Deakin Applied Artificial Intelligence Initiative
Deakin University
221 Burwood Highway
Burwood, VIC 3125

Dear Associate Professor Scott Barnett and Stefanus Kurniawan,

RE: Application for the Curating Real-World Manufacturing Evaluation Datasets research internship

I am writing to apply for the **Curating Real-World Manufacturing Evaluation Datasets to Assess the AI Models and Agentic Harnesses behind Overseer** opportunity, advertised through Deakin's **High Achievers' Internship Program with the Applied Artificial Intelligence Initiative**. What drew me to this project is a question I have started asking about my own use of AI: when an AI's answer seems to work, how do you know it is actually right, and how would you notice if it quietly got worse?

I am completing a Bachelor of Software Engineering (Honours), expected to graduate in 2028, with an 81.33 WAM and scores of 100 in Backend Development and Data Structures and Algorithms. I came up with the idea for ParkSense, a Melbourne parking-availability predictor that won first place in a Deakin Data Science Club hackathon. My most important contribution came before the build: finding usable public data sources, working out what the City of Melbourne sensor feed actually measured, and judging whether it carried enough real signal to predict from. I later contributed to the backend foundations and reviewed pull requests on the team's shared codebase, which taught me to extend software I had not written alone.

I have also learnt to make systems prove their results. My backend project, which scored 100/100 and was showcased at Deakin Open Day, controls a physical robot I built. When the robot reconnects and reports what it did, the server replays those actions through a simulation and only saves them if they check out, so it can never record the robot somewhere it never was. On the Deakin Software Engineering Club committee, I designed the bugs seeded into a workshop app for around 30 students, each clear enough that nobody could argue about whether it had been fixed. That is reflects the checkable success criteria your brief asks for.

What excites me about Overseer is that it already runs on real factory processes with real industry partners. I use Claude Code regularly, and my usual test is still whether its output appears to work. That tells me very little about whether the model has improved, regressed on one task, or simply produced a plausible result. I have an impression of its reliability, not a measurement. A factory cannot run on an impression. I am curious how you decide the right answer for a factory task that seems to need judgement, and how a dataset can catch a model getting worse at one task a partner relies on while its overall score still looks fine.

I would bring careful work with messy data, a habit of making systems prove their results, and experience working in codebases I did not start. I am available on campus at Burwood two days per week for the full ten weeks. Please find my resume and academic transcript attached. I would welcome the opportunity to discuss my application in an interview. Thank you for your consideration.

Kind regards,

Manit Khera